

Shuchang Xu

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Research Interests

Human Computer-Interaction, Accessibility, Multi-Modal Machine Learning, Video & Game Interaction

Education

The Hong Kong University of Science and Technology	Hong Kong, China
Ph.D. in Computer Science, Advisor: Dr. Huamin Qu	Aug 2023 - Present
Tsinghua University	Beijing, China
M.S. in Computer Science, Advisor: Dr. Yuanchun Shi	Aug 2018 - Jun 2021
Tsinghua University	Beijing, China
B.S. in Electrical Engineering, GPA: 91.2/100 (Top 5%)	Aug 2014 - Jun 2018
B.S. in Digital Media Art and Design (Second Major)	
University of Calgary	Calgary, Canada
Visiting Student, Advisor: Dr. Laleh Behjat	Jun 2017 - Sep 2017

Work Experience

Xiaomi Corporation	Beijing, China
Software Product Manager, Smartphone Division	Aug 2021 - Jun 2023

Publications

- [1] **DanmuA11y: Making Time-Synced On-Screen Video Comments (Danmu) Accessible to Blind and Low Vision Users via Multi-Viewer Audio Discussions.**
Shuchang Xu, Xiaofu Jin, Huamin Qu, Yukang Yan.
CHI 25 - ACM Conference on Human Factors in Computing Systems.
- [2] **Memory Reviver: Supporting Photo-Collection Reminiscence for People with Visual Impairment via a Proactive Chatbot.**
Shuchang Xu, Chang Chen, Zichen Liu, Xiaofu Jin, Linping Yuan, Yukang Yan, Huamin Qu.
UIST 2024 - ACM Symposium on User Interface Software and Technology.
- [3] **Virtual Paving: Rendering a Smooth Path for People with Visual Impairment through Vibrotactile and Audio Feedback.**
Shuchang Xu, Ciyuan Yang, Wenhao Ge, Chun Yu, Yuanchun Shi.
IMWUT 2020 - Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies.
- [4] **LightGuide: Directing Visually Impaired People along a Path Using Light Cues.**
Ciyuan Yang, Shuchang Xu (co-first), Tianyu Yu, Guanhong Liu, Chun Yu, Yuanchun Shi.
IMWUT 2021 - Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies.
- [5] **Tactile Compass: Enabling Visually Impaired People to Follow a Path with Continuous Directional Feedback.**
Guanhong Liu, Tianyu Yu, Chun Yu, Haiqing Xu, Shuchang Xu, Ciyuan Yang, Feng Wang, Haipeng Mi, Yuanchun Shi.
CHI 2021 - ACM Conference on Human Factors in Computing Systems.
- [6] **Accurate and Low-Latency Sensing of Touch Contact on Any Surface with Finger-Worn IMU Sensor.**

Yizheng Gu, Chun Yu, Zhipeng Li, Weiqi Li, **Shuchang Xu**, Xiaoying Wei, Yuanchun Shi.
UIST 2019 - ACM Symposium on User Interface Software and Technology.

- [7] **A Machine Learning Framework to Identify Detailed Routing Short Violations from a Placed Netlist.**

Aysa Tabrizi, Nima Darav, **Shuchang Xu**, Logan Rakai, Ismail Bustany, Andrew Kennings, Laleh Behjat.
DAC 2018 - ACM/IEEE Design Automation Conference.

Fellowship & Awards - Ph.D.

Hong Kong Postgraduate Fellowship	2024–2027
Full stipend, approximately USD 55,000 per year	
Best Paper Honorable Mention Award	Apr. 2025
CHI 2025, among the top 5% out of 5020 submissions	
HKUST RedBird Academic Excellence Award	2024–2025
HKD 40,000 for outstanding research performance	
Research Travel Awards	2024–2025
HKD 14,000 for CHI 2025; HKD 14,700 for UIST 2024	
Special Recognition for Outstanding Reviews	
CHI 2025, IMWUT 2025, ISMAR 2025, CHI 2025 LBW	

Fellowship & Awards - Master & Undergraduate

Outstanding Graduates in Computer Science, Tsinghua University	Jun 2021
China National Scholarship for Postgraduates (Top 1%)	Nov 2020
China National Scholarship for Postgraduates (Top 1%)	Nov 2019
China National Scholarship for Undergraduates (Top 1%)	Nov 2017
Best Paper Award for HCI in Games, HCII 2019	Jul 2019
Xiaomi Artificial Intelligence Hackathon, 2nd Prize	Jul 2022
Microsoft Mixed Reality Design Contest, 2nd Prize	Dec 2017

Invited Talks

UIUC SALT Lab , Invited by Dr. Yun Huang	Jun 2025
Improving Visual Media Accessibility with AI-Assisted Interactive Storytelling Techniques	
Tongji University , Invited by Dr. Guanhong Liu	Apr 2025
Multi-Modal AI Systems for Improving Visual Media Accessibility	

Teaching Experiences

Student Research Mentor , UX Design, Tongji University	2025
Teaching Assistant , Java Programming Language, HKUST	2025
Teaching Assistant , UX Design, Tsinghua University	2020
Teaching Assistant , Big Data and Machine Intelligence, Tsinghua University	2019

Patents

Vision Accessibility Techniques: CN111494175
Mobile Game Interaction Techniques: CN109316745, CN109460179, CN109550242
Smartphone Interaction Techniques: CN115857375, CN115755635

Professional Experiences in Creative Media

Bytedance – TikTok Technical Artist Intern – Shanghai, China	Jun 2020 - Aug 2020
Tencent – Game Level Designer Intern – Shenzhen, China	Jun 2019 - Aug 2019
NetEase – Game Interaction Designer Intern – Hangzhou, China	Jun 2018 - Aug 2018

Skills

Programming: Swift, C#, Java, Python, React Native, C++, OpenGL, OpenCV

Prototyping: Hardware (Arduino, Rasperry PI); 3D (3ds Max, Maya), UX Design (After Effects, Figma)

Platforms/IDE: Unity3D, Xcode, Android Studio

TOEFL: 109/120 (R 29/30, L 26/30, S 24/30, W 30/30)